

Association of Aberrations in One-Carbon Metabolism With Molecular Phenotype and Grade of Breast Cancer

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We have earlier demonstrated the role of aberrant one-carbon metabolism in the etiology of breast cancer. In the current study, we examine the clinical utility of these factors in predicting the subtype of breast cancer and as indicators of disease progression. Polymerase chain reaction (PCR)–restriction fragment length polymorphism (RFLP) and PCR–amplified fragment length polymorphism (AFLP) approaches were used for genetic analysis. Plasma folate and homocysteine were measured using AxSYM folate kit and reverse phase HPLC, respectively. Multiple linear regression models were used to test the predictability of disease progression. Luminal A subtype was associated with late age of onset, higher body mass index and lack of family history of breast cancer. Thymidylate synthase (TYMS) 5'-UTR 28 bp tandem repeat (OR: 2.09, 95% CI: 1.05–4.16) and methylene tetrahydrofolate reductase (MTHFR) C677T (OR: 4.10, 95% CI: 1.40–11.95) were strongly associated with Luminal B. Reduced folate carrier (RFC1) G80A (OR: 2.92, 95% CI: 1.22–6.97) and methionine synthase (MTR) A2756G (OR: 4.71, 95% CI: 1.66–13.31) polymorphisms were associated with LuminA-HH subtype while MTHFR C677T showed association with HER-enriched (OR: 30.41, 95% CI: 6.47–142.91). Cytosolic serine hydroxymethyltransferase (cSHMT) conferred protection against basal-like breast cancer (OR: 0.47, 95% CI: 0.22–0.98). HER-enriched and basal-like subtypes showed positive association with familial breast cancer and inverse association with plasma folate. Hyperhomocysteinemia was observed in Luminal B and basal-like subtypes. Multiple linear regression models of aberrant one-carbon metabolism were found to be moderate predictors of breast cancer grade (area under the receiver operating characteristic curve, $C = 0.72$, 95% CI: 0.58–0.87, $P = 0.008$). To conclude, aberrations in one-carbon metabolism predict the subtype of breast cancer and disease progression. © 2011 Wiley Periodicals, Inc.

Key words: breast cancer; phenotype; one-carbon metabolism; polymorphism; plasma folate; plasma homocysteine; estrogen receptor (ER); progesterone receptor (PR); human epidermal growth factor receptor 2 (HER2/Neu)

INTRODUCTION

Although the clinicopathological heterogeneity of breast cancer is well documented, little is known about the genes that influence the phenotype of breast cancer. The highly penetrant mutations in BRCA1 and BRCA2 account for small proportion of breast cancer cases [1]. These mutations were often associated with poor tumor differentiation, positive lymph node invasion, vascular peritumoral invasion [2], decreased expression of estrogen receptor (ER), progesterone receptor (PR), human epidermal growth factor receptor 2 (HER2/Neu), and increased expression of p53 [3]. Certain breast cancer cases mimic this phenotype despite having wild BRCA1 due to low-level promoter methylation of BRCA1 [4]. Basal-like cancers were reported to show infrequent CCND1 amplification and cyclin D1 overexpression [5]. Genomic instability i.e., loss of heterozygosity (LOH)/allelic

imbalance on chromosome (chr) 11 and at chr 1, 2, 5, 18, 20, and 22 was found to be associated with the tumor grade and regional lymph node metastasis, respectively [6]. There was a positive

Abbreviations: PCR, polymerase chain reaction; AFLP, amplified fragment length polymorphism; RFLP, restriction fragment length polymorphism; TYMS, thymidylate synthase; MTHFR, methylenetetrahydrofolate reductase; RFC1, reduced folate carrier 1; MTR, methionine synthase; cSHMT, cytosolic serine hydroxymethyltransferase; ER, estrogen receptor; PR, progesterone receptor; HER2/Neu, human epidermal growth factor receptor 2; THF, tetrahydrofolate; SAM, S-adenosyl methionine; GCPII, glutamate carboxypeptidase II; BMI, body mass index; ROC, receiver operating characteristic; MTRR, methionine synthase reductase.

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correlation between the LOH at Chr 14 in the epithelium with the PR expression, specifically in sporadic primary invasive breast carcinoma [6]. Differential pattern of alterations (deletions and amplifications) were observed in early onset and late onset breast cancers. In younger group, the sequence being BRCA1 > TP53 > ATM > BRCA2 > RB1 > Cyclin D1 > BRCAX, while that of in the older group being TP53 > RB1 > BRCA1 > ATM > BRCA2 > Cyclin D1 > BRCAX [7].

Since majority of the breast cancer cases are sporadic, there is a likelihood of involvement of less penetrant genetic polymorphisms encoding diverse range of proteins that can interact to define a high-risk phenotype. The association of putatively functional polymorphisms in genes regulating synthesis, repair and methylation of DNA; steroid hormone metabolism and signaling; and carcinogen metabolism on breast cancer survival have generated lot of curiosity on these factors as the possible prognostic markers [8].

Genomic instability and altered gene expression, the two important aspects that were observed to be associated breast cancer phenotype, are likely to be modulated by one-carbon metabolism. This metabolic pathway, as a result of sequential transfer of one-carbon moiety from one substrate to another, forms several crucial metabolic intermediates such as formyl tetrahydrofolate (formyl THF), 5,10-methylene THF, 5-methyl THF, methionine and S-adenosyl methionine (SAM). These are crucial for maintaining one-carbon homeostasis, thymidylate synthesis, remethylation of homocysteine to methionine, synthesis of SAM and methylation of DNA, proteins, and catecholamines, respectively. Aberrations in one-carbon metabolism adversely affect the synthesis, repair and methylation of DNA, thereby increasing the risk for breast cancer [9].

Several researchers have explored the role of folate intake in modulating breast cancer risk [10–14]. Lee et al. [10] have observed decreased risk for breast cancer in women who consume green vegetables, white vegetables, mushrooms, and increased risk in women who consume meat. Chen et al. [11] showed greater reduction in the breast cancer risk among non-supplement users in the highest quintile of dietary folate intake as compared with non-supplement users in the lowest quintile of dietary folate intake. Sharp et al. [12] also reported reduced risk for breast cancer among women with highest dietary intake of folate. Chou et al. [13] have observed inverse association between plasma folate and breast cancer. Beilby et al. [14] have observed lower serum folate levels in breast cancer cases than controls (5.7 vs. 6.6 µg/L). Global hypomethylation and focal hypermethylation, which are hallmarks of cancer [15], are in turn influenced by one-carbon metabolism. Global

hypomethylation specifically at heterochromatin activates proto-oncogenes, while focal hypermethylation at promoter CpG islands of tumor suppressor genes inactivate the tumor suppressor function thus increasing the risk for cancer. A recent study showed association of tumor grade, size, estrogen, and PR status, and triple negative status with altered methylation of 209, 74, 183, 69, and 130 autosomal loci, respectively, with dietary folate being important modulator of methylation status [16]. However, when limited number of loci such as the promoter methylation of E-cadherin, p16, RAR-beta2 were explored, no association was observed with aberrations in one-carbon metabolism i.e., folate intake, methylene tetrahydrofolate reductase (MTHFR) C677T, and methionine synthase (MTR) A2756G [17]. The ER, PR, HER2/Neu status, which is widely accepted as the prognostic marker and indicator of tailoring breast cancer treatment [18], might also be altered by these aberrations as a result of modifications in the methylation profiles of these genes.

There are limited reports correlating the clinicopathological characteristics of breast cancer with the one-carbon metabolism [19,20]. Studies showed no association between the clinicopathological characteristics of breast cancer and polymorphisms in thymidylate synthase (TYMS 28 bp tandem repeat), MTHFR C677T [17], cystathionine beta synthase (CBS 68 bp ins), DNA methyl transferase 3B (DNMT3B C46359T), methylene tetrahydrofolate dehydrogenase (MTHFD1 G1958A), MTR A2756G [20]. An inverse association between the MTR A2756G and ER/PR status; and positive association between the frequency of CpG island promoter methylation of the RARβ2, CDH1, ER, BRCA1, CCND2, p16, and TWIST genes and TYMS 28 bp tandem repeat/MTHFD1 G1958A polymorphisms were reported [20]. Till date, there are no genotype-phenotype association studies on polymorphisms in glutamate carboxypeptidase II (GCP2) C1561T, reduced folate carrier 1 (RFC1) G80A and cytosolic serine hydroxymethyl transferase (cSHMT C1420T) that can influence absorption and transport of folate; and synthesis of 5,10-methylene tetrahydrofolate.

Recently, we have explored the role of aberrations in one-carbon metabolism in altering the susceptibility to breast cancer and observed strong epistatic i.e., gene-gene interactions between RFC1 G80A, TYMS 5'-UTR 28 bp tandem repeat and MTHFR C677T variants [21]. Further, one polymorphism i.e., cSHMT C1420T was found to confer protection by increasing the folate pool [21]. In another recent study, we have demonstrated that RFC1 G80A, elevated plasma homocysteine and folate deficiency induce elevated 8-oxo-2-deoxyguanosine, a mutagenic lesion in DNA that triggers carcinogenesis [22]. Further, protective role of

increased folate and cSHMT C1420T polymorphism was also demonstrated [22]. We hypothesize that the aberrations of one-carbon metabolism may also be associated with the breast cancer phenotype either by affecting synthesis or methylation of DNA. In the current study, we evaluated the association between the clinicopathological characteristics of breast cancer with the eight genetic polymorphisms in one-carbon metabolisms, plasma folate, and plasma homocysteine.

MATERIALS AND METHODS

Subjects

All the breast cancer cases that were diagnosed based on the mammography and histopathological findings were enrolled in the study at Department of medical and surgical oncology, Nizam's Institute of Medical Sciences (NIMS), Hyderabad, Andhra Pradesh, India during the period of January 2009 to January 2011. Sample size calculation was done based on the findings of the pilot study on MTHFR C677T polymorphism (T-allele frequency in cases and controls: 0.13 vs. 0.08), which indicated 828 alleles (414 subjects) to be tested to attain 90% power with type 1 error of 0.05. Using this as a baseline, 558 subjects (305 breast cancer cases and 253 healthy female controls) were enrolled for the study. Out of which 19 subjects were excluded due to lack of access to clinical information or medical records. Finally, 286 breast cancer cases and 253 healthy controls (539 subjects) were available to study the genotype-phenotype correlation. Controls were healthy volunteers, mainly hospital staff or their relatives with no history of any malignant disease or benign breast disease.

Demographic Factors

During the in-person baseline interview, subjects were asked detailed information on age, ethnic origin, food frequency (based on standardized food frequency questionnaire), personal habits such as exercise, smoking, alcohol intake, tobacco chewing, reproductive history such as age of menarche, age at the time of first full term pregnancy, number of live births, number of miscarriages, history of breast feeding, pre/post menopausal status, family history of breast cancer or any other cancer etc. Height and weight were measured to calculate body mass index (BMI, kg/m²). All the personal interviews were conducted during the first visit to oncologist after the confirmed diagnosis of breast cancer (for cases) and samples were collected followed by the interview to avoid influence of post-surgical trauma or radiation/chemotherapy. This study was approved by the Institutional Ethical Committee of Nizam's Institute of Medical

Table 1. Immunohistochemical Subtyping Criteria for Breast Cancer Tumors

	ER	PR	HER	Ki67
Luminal A	++	++	–	<14%
Luminal B	+	+	–	>14%
LuminA-HH	++	++	+	<14%
LuminB-HH	+	+	+	>14%
HER-enriched	–	–	+	
Basal-like	–	–	–	

ER, estrogen receptor; PR, progesterone receptor; HER2/Neu, human epidermal growth factor receptor 2; Ki67, a cellular marker of proliferation.

Sciences, Hyderabad, India. Informed consent was obtained from each subject.

Immunohistochemistry and Fluorescent In Situ Hybridization

Immunohistochemistry for ER, PR, HER2, and Ki67 was performed concurrently on serial sections with the standard streptavidin-biotin complex method with 3,3'-diaminobenzidine as the chromogen. ER (Clone SP1; Lab Vision) antibody was used at 1:250 dilution with an 8-min microwave antigen retrieval in a 10 mmol/L (pH 6.0) citrate buffer. Ready-to-use PR (Clone 1E2; Ventana, Tucson AZ) antibody was used following the Ventana automated stainer standard CC1 protocol. HER2 staining was done using HER2 (Clone SP3; Lab Vision) antibody at 1:100 dilution with heat-induced antigen retrieval using 0.05 mol/L Tris buffer (pH 10.0) heated to 95°C in a steamer for 30 min. The Ki67 antibody (clone SP6; Thermo Scientific, Fremont, CA) was applied at a 1:200 dilution for 32 min, by following the Ventana Benchmark automated immunostainer (Ventana) standard Cell Conditioner 1 (CC1, a proprietary buffer) protocol at 98°C for 30 min. For HER2 fluorescent in situ hybridization assay, slides were hybridized with probes to LSI (locus-specific identifier), HER2/Neu and to centromere 17 by use of the PathVysion HER-2 DNA Probe kit (Abbott Molecular, Abbott Park, IL) according to manufacturer's instructions, with modifications to pretreatment and hybridization. Slides were counterstained with 4, 6-diamidino-2-phenylindole, stained material was visualized on a Zeiss Axioplan epifluorescent microscope, and signals were analyzed with a Metafer image acquisition system (Metasystems, Altlussheim, Germany) [23].

Biomarker expression from immunohistochemistry assays was scored by two surgical pathologists, who were blinded to the clinicopathological characteristics. Tumors were considered positive for ER or PR if immunostaining was observed in more than 1% of tumor nuclei. Tumors were considered positive for HER2 if immunostaining was scored as

3+ according to HercepTest criteria, with an amplification ratio for fluorescent in situ hybridization of 2.0 or more being the cut point that was used to segregate immunohistochemistry equivocal tumors (scored as 2+). Ki67 was visually scored for percentage of tumor cell nuclei with positive immunostaining above the background level by two pathologists. The criteria used for immunohistochemical subtyping were depicted in Table 1. Tumor grading was done according to the Scarff-Bloom-Richardson (SBR) scale and the lymph node involvement was assessed by histopathological examination. Briefly, the grading was done by combining tubule formation (>75%: 1, 10–75%: 2, and <10%: 3), nuclear pleomorphism (small regular uniform cells: 1, moderate nuclear size and variation: 2, and marked nuclear variation: 3) and mitoses/10 hpf (0–9: 1, 10–19: 2, and ≥20: 3). Tumors grading was done based on cumulative histological grade [low grade (I): 3–5, intermediate grade (II): 6–7, and high grade (III): 8–9]. The clinicopathological characteristics of studied population are represented as Table 3.

Sample Collection

Whole blood samples were collected from all the eligible subjects in two ethylenediamine tetraacetic

acid (EDTA) vacutainers. Plasma samples were separated immediately following centrifugation at 3,300g/10 min at 4°C and stored in aliquots at –70°C until analysis.

Estimation of Biochemical Parameters

Total plasma homocysteine was determined by using reverse phase HPLC method [24]. Plasma folate levels were estimated using AxSYM folate kit (Abbott Laboratories). Subjects who are on any vitamin or antioxidant supplements and patients who had radiation and/or chemotherapy were excluded from this analysis.

Genetic Analysis

Genomic DNA was isolated from the buffycoat using standard protocols. Polymerase chain reaction (PCR) was carried out to amplify the target DNA and the restriction fragment length polymorphism (RFLP) and amplified fragment length polymorphism (AFLP) techniques were used for analyzing the common polymorphisms in genes regulating one-carbon metabolism [22] (Table 2).

Each PCR set was accompanied with a negative control without genomic DNA, known homozygote or known heterozygote to check the quality of PCR and restriction digestion. Furthermore, all

Table 2. Reaction Conditions for PCR-Based Genetic Analysis

Polymorphism	Primers (5'–3')	PCR conditions	Restriction enzyme	Restriction site
GCP II C1561T (×35)	CAT TCT GGT AGG AAT TTA GCA AAA CAC CAC CTA TGT TTA ACA	D: 95°C/30 s A: 50°C/30 s E: 72°C/ 30s	AccI	+ (T allele)
RFC G80A (×30)	AGT GTC ACC TTC GTC CCC TC CTC CCG CGT GAA GTT CTT-	D: 95°C/1 min A: 59°C/1 min E: 72°C/1 min	HhaI	– (A allele)
cSHMT C1420T (×35)	GTG TGG GGT GAC TTC ATT TGT G GGA GCA GCT CAT CCA TCT CTC	D: 95°C/30 s A: 56°C/30 s E: 72°C/30 s	EcoRI	+ (C allele)
TYMS 5'-UTR (AFLP) (×35)	CGT GGC TCC TGC GTT TCC GAG CCG GCC ACA GGC AT	D: 95°C/30 s A: 62°C/30 s E: 72°C/45 s	AFLP	AFLP
TYMS 3'-UTR Ins6/del6 (×30)	CAAATCTGAGGGAGCTGAGT CAGATAAGTGGCAGTACAGA	D: 95°C/30 s A: 58°C/45 s E: 72°C/45 s	DraI	– (del6)
MTHFR C677T (×30)	TTT GAG GCT GAC CTG AAG CAC TTG AAG GAG GAG TGG TAG CCC TGG ATG GGA AAG ATC CCG	D: 95°C/1 min A: 60°C/1 min E: 72°C/1 min	HinfI	+ (T allele)
MTR A2756G (×40)	TGT TCC CAG CTG TTA GAT GAA AAT C GAT CCA AAG CCT TTT ACA CTC CTC	D: 95°C/30 s A: 60°C/30 s E: 72°C/30 s	HaeIII	+ (G allele)
MTRR A66G (×30)	GCA AAG GCC ATC GCA GAA GAC AT GTG AAG ATC TGC AGA AAA TCC ATG TA	D: 95°C/1 min A: 55°C/30 s E: 72°C/30 s	NdeI	+ (A allele)

D, denaturation; A, annealing; E, extension; ×, number of cycles; +, restriction site created; –, restriction site abolished; GCP II, glutamate carboxypeptidase II; RFC1, reduced folate carrier 1; cSHMT, cytosolic serine hydroxymethyltransferase; TYMS, thymidylate synthase; MTHFR, methylene tetrahydrofolate reductase; MTR, methionine synthase; MTRR, methionine synthase reductase.

the genotypes were checked for Hardy–Weinberg equilibrium as the quality control.

Statistical Analysis

Student's *t*-test was used to evaluate the differences in continuous variables. Allele frequencies for each polymorphism were calculated and the χ^2 -test was used to test whether the distribution of polymorphisms is in accordance with Hardy–Weinberg equilibrium. All the genotypes were computed as 0, 1, and 2 to depict wild, heterozygous and homozygous mutant genotypes. An additive model was adopted to obtain odds ratios. Fisher's exact test was used to evaluate the possible association between the genetic variants and various clinicopathological characteristics by computing the data in 2×2 contingency table format. Since the majority of polymorphisms have limited number of homozygous mutant subjects, variant allele frequency was used to study the main effect. Multiple logistic regression analysis was employed to test independent associations after controlling for confounding physiological (age, BMI, age of menarche, age at first full term pregnancy, parity, and menopausal status) and other genetic variables. For the receiver operating characteristic (ROC) curve analysis, we have selected the polymorphisms and physiological variables that showed statistically significant correlation coefficients. An algorithm based on multiple linear regression model was developed that can predict the tumor grade. The sensitivity and specificity of prediction based on these factors was used for ROC curve plotting. All the "*P*" values are derived from two-tailed statistical tests. All the analyses were performed with the statpages.org statistical web package.

RESULTS

The distribution of clinicopathological characteristics such as histological subtype, breast cancer grade, axillary lymph node involvement, immunohistochemical phenotypes is shown in Table 3. Ductal involvement was observed in 97.2% breast cancer cases. Lobular, mixed (ductal and lobular), and mucinous accounting for remaining cases. Low grade tumors were observed in 19.9% cases while intermediate grade and high grade tumors were observed in 39.2% and 40.9% cases, respectively. Axillary lymph node involvement was observed in 38.8% cases. HER2-negative breast tumors accounted for 65.7% cases, which were subdivided into luminal A, luminal B, and basal-like subtypes based on ER, PR, and Ki67 expression. HER2-positive breast tumors accounted for 34.3% cases, which were subdivided into luminA-HH, luminB-HH, and HER2-enriched based on ER, PR, and Ki67 expression.

Table 3. Clinicopathological Characteristics of Enrolled Breast Cancer Cases

Clinicopathological characteristic	Number of subjects
Histological subtype	
Ductal	278
Lobular	4
Mixed (ductal and lobular)	3
Mucinous	1
Breast cancer grade (SBR scale)	
Low grade (stage I)	57
Intermediate grade (stage IIa & IIb)	112
High grade (stage IIIa, IIIb, & IV)	117
Axillary lymph node involvement	
Yes	111
No	173
Distant metastasis	2
Immunohistochemical phenotypes	
Luminal A	92
Luminal B	46
Basal-like	50
LuminA-HH	36
LuminB-HH	26
HER2-enriched	36

Table 4 represents the distribution of different genotypes across different breast cancer phenotypes. As depicted in Table 5, characteristic patterns were observed in demographic and genetic risk factor profile among luminal A, luminal B, and basal-like subtypes. The risk for luminal A subtype was found to increase with age and BMI; and decrease with the onset of menopause. TYMS 5'-UTR 2R3R and MTHFR C677T polymorphisms showed 2.09- (95% CI: 1.05–4.16) and 4.10- (95% CI: 1.40–11.95) folds increased risk for luminal B subtype. Familial breast cancer was associated with basal-like subtype. Post-menopausal women and women with cSHMT C1420T polymorphism were less likely to have basal-like subtype.

As shown in Table 6, RFC1 G80A and MTR A2756G polymorphisms were associated with 2.92- (95% CI: 1.22–6.97) and 4.71- (95% CI: 1.66–13.31) folds risk for luminA-HH subtype. None of the demographic or genetic factors were found to be associated luminB-HH subtype. The HER2-enriched subtype showed association with familial breast cancer and MTHFR C677T polymorphism.

Further, we evaluated the impact of plasma folate and plasma homocysteine on breast cancer subtypes (Table 7). Low plasma folate was associated with HER-2 enriched and basal-like subtypes. Luminal B and basal-like subtypes showed elevated homocysteine levels.

As depicted in Table 8, tumor grading based on tubule formation and nuclear pleomorphism and mitotic index was found to be greatly associated by demographic and genetic factors. Tubule

Table 4. Distribution of Genotypes Across Different IHC Phenotypes

Variable	Luminal-A (n = 92)			Luminal B (n = 46)			Basal-like (n = 50)			LuminA-HH (n = 36)			LuminB-HH (n = 26)			HER-enriched (n = 36)			Controls (n = 253)		
	N1	N2	N3	N1	N2	N3	N1	N2	N3	N1	N2	N3	N1	N2	N3	N1	N2	N3	N1	N2	N3
GCPII C1561T	78	14	0	40	6	0	44	6	0	30	6	0	22	4	0	32	4	0	203	50	0
RFC1 G80A	38	40	14	12	28	6	18	24	8	12	11	14	10	10	6	14	14	8	99	137	17
SHMT C1420T	16	42	34	6	26	14	14	20	16	6	16	14	2	24	2	8	16	12	24	154	75
TYMS 5' 2R3R	40	28	24	14	22	10	22	22	6	14	22	0	12	11	4	16	14	6	111	116	26
TYMS 3' Ins6/del6	32	38	22	12	14	20	8	26	16	8	22	6	12	11	4	8	22	6	59	136	58
MTHFR C677T	74	18	0	32	14	0	40	10	0	28	6	2	24	2	0	18	16	2	222	31	0
MTR A2756G	44	48	0	22	18	6	16	34	0	4	26	6	12	11	4	22	14	0	126	117	20
MTRR A66G	26	66	0	10	32	4	12	34	4	6	30	0	6	20	0	10	24	2	80	161	12

N1, N2, and N3 represent wild, heterozygous and homozygous mutant genotypes; bold genotypes showed statistically significant difference with controls.

formation showed positive association with familial breast cancer, RFC1 G80A and MTR A2756G polymorphisms; and inverse association with age at menarche, cSHMT C1420T, TYMS 5'-UTR 2R3R and MTHFR C677T polymorphisms. Multiple linear regression models of aberrations in one-carbon metabolism were found to be moderate predictors of breast cancer grade (area under the ROC curve, C = 0.72, 95% CI: 0.58–0.87, P = 0.008; Figure 1).

DISCUSSION

In view of clinical and immunohistochemical heterogeneity in breast cancer presentation, it can be presumed that molecular alterations might be contributing to phenotypic variation. Since the high penetrant gene mutations account for the small proportion of cases, they are insufficient to explain these variations. The gene expression markers and DNA methylation profiles that were developed recently can be utilized as prognostic indicators of breast cancer [16]. We have hypothesized that any aberration in the one-carbon metabolism, the key player in genome integrity and DNA methylation, might be associated clinical or IHC phenotypic variability of breast cancer.

Among the demographic characteristics, age, BMI, menopause, and family history of breast cancer were found to differ among different subtypes of breast cancer consistent with the observations by Yang and co-workers. [25]. In the current study, Age, BMI, and pre-menopausal status were associated with luminal A subtype. Basal-like subtype was associated with familial breast cancer and pre-menopausal status. HER2-enriched was associated with familial breast cancer.

The hormone receptor status, another prognostic indicator, was observed to be associated by the genetic factors, plasma folate, and plasma homocysteine. TYMS 5'-UTR 2R3R and MTHFR C677T variant alleles were associated with luminal B subtype. RFC1 G80A and MTR A2756G variant alleles were associated with luminA-HH subtype. The Her2-enriched subtype showed association with MTHFR C677T variant allele. Our study is in agreement with Li et al. [20] in demonstrating null association between MTHFR C677T and ER/PR status. Larsson et al. [26] have reported an inverse association between the folate intake and ER + / PR-tumors. Furthermore, Maruti et al. [27] have demonstrated the potential benefit of high dietary intake of folate, particularly against ER-negative breast cancers. The association of HER-2 enriched and basal-like subtypes with low plasma folate and inverse association of cSHMT C1420T variant allele with basal-like subtype can be explained based on the observations by our group [21] and Heil et al. [28] that cSHMT C1420T variant allele increases plasma folate pool. Elevated plasma homocysteine

Table 5. Association of Demographic and Genetic Factors With HER-Negative Breast Tumors ($n = 188$)

Variable	Adjusted OR (95% CI)		
	Luminal A ($n = 92$)	Luminal B ($n = 46$)	Basal-like ($n = 50$)
Age	1.09 (1.04–1.15)****	1.00 (0.94–1.07)	1.06 (1.00–1.14)
BMI	1.08 (1.02–1.14)**	0.98 (0.87–1.10)	0.92 (0.82–1.04)
Age at menarche	0.97 (0.73–1.28)	0.90 (0.63–1.29)	1.08 (0.75–1.55)
Parity	0.82 (0.62–1.09)	1.05 (0.73–1.50)	0.89 (0.63–1.27)
Menopause	0.12 (0.04–0.39)***	1.40 (0.29–6.70)	0.21 (0.04–1.00)*
F/H of breast cancer	0.00	3.18 (0.28–36.33)	10.24 (1.86–56.50)**
GCP11 C1561T	0.97 (0.35–2.71)	0.68 (0.17–2.65)	0.75 (0.18–3.07)
RFC1 G80A	1.47 (0.79–2.74)	1.91 (0.85–4.27)	1.60 (0.75–3.41)
cSHMT C1420T	1.17 (0.64–2.15)	1.05 (0.46–2.38)	0.47 (0.22–0.98)*
TYMS 5'-UTR 2R3R	1.51 (0.89–2.56)	2.09 (1.05–4.16)*	1.11 (0.55–2.27)
TYMS 3'-UTR del6/ins6	0.81 (0.47–1.40)	1.52 (0.73–3.17)	1.21 (0.58–2.53)
MTHFR C677T	2.40 (0.92–6.29)	4.10 (1.40–11.95)**	2.21 (0.69–7.09)
MTR A2756G	0.85 (0.42–1.69)	1.34 (0.61–2.94)	1.55 (0.67–3.54)
MTRR A66G	1.46 (0.71–3.03)	1.39 (0.55–3.52)	1.57 (0.66–3.74)

Adjusted OR, odds ratio adjusted for other demographic and genetic factors depicted in the table; CI, confidence interval; n = number of subjects; F/H, family history; GCP11, glutamate carboxypeptidase II; RFC1, reduced folate carrier 1; cSHMT, cytosolic serine hydroxymethyltransferase; TYMS, thymidylate synthase; MTHFR, methylene tetrahydrofolate reductase; MTR, methionine synthase; MTRR, methionine synthase reductase.

An additive genetic model was used to estimate odds ratios.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

**** $P < 0.0001$.

Table 6. Association of Demographic and Genetic Factors With HER-Positive Breast Tumors ($n = 98$)

Variable	Adjusted OR (95% CI)		
	LuminA-HH ($n = 36$)	LuminB-HH ($n = 26$)	HER2-enriched ($n = 36$)
Age	1.06 (0.99–1.14)	1.00 (0.92–1.09)	1.07 (0.98–1.16)
BMI	1.08 (0.98–1.18)	1.08 (0.93–1.24)	1.03 (0.85–1.24)
Age at menarche	1.28 (0.85–1.94)	1.01 (0.60–1.71)	1.04 (0.62–1.76)
Parity	0.88 (0.58–1.35)	0.76 (0.42–1.35)	0.93 (0.59–1.46)
Menopause	0.20 (0.03–1.20)	3.72 (0.36–38.18)	1.16 (0.11–11.87)
F/H of breast cancer	0.00	8.39 (0.40–177.40)	20.75 (1.39–310.41)*
GCP11 C1561T	0.78 (0.17–3.53)	0.40 (0.05–2.99)	1.05 (0.15–7.50)
RFC1 G80A	2.92 (1.22–6.97)*	1.88 (0.57–6.20)	2.56 (0.85–7.70)
cSHMT C1420T	1.02 (0.41–2.55)	0.61 (0.19–1.96)	1.63 (0.50–5.27)
TYMS 5'-UTR 2R3R	1.02 (0.41–2.55)	1.03 (0.38–2.79)	1.71 (0.65–4.54)
TYMS 3'-UTR del6/ins6	0.87 (0.33–2.28)	0.46 (0.16–1.32)	0.40 (0.14–1.10)
MTHFR C677T	0.75 (0.31–1.78)	0.56 (0.05–6.84)	30.41 (6.47–142.91)****
MTR A2756G	4.71 (1.66–13.31)***	1.54 (0.48–4.92)	0.40 (0.11–1.46)
MTRR A66G	1.64 (0.52–5.14)	1.16 (0.36–3.74)	1.88 (0.54–6.55)

Adjusted OR, odds ratio adjusted for other demographic and genetic factors depicted in the table; CI, confidence interval; n = number of subjects; GCP11, glutamate carboxypeptidase II; RFC1, reduced folate carrier 1; cSHMT, cytosolic serine hydroxymethyltransferase; TYMS, thymidylate synthase; MTHFR, methylene tetrahydrofolate reductase; MTR, methionine synthase; MTRR, methionine synthase reductase.

An additive genetic model was used to estimate odds ratios.

* $P < 0.05$.

*** $P < 0.005$.

**** $P < 0.0001$.

Table 7. Association of Plasma Folate and Plasma Homocysteine and Breast Cancer Subtypes

Study group	Plasma folate (ng/ml)		Plasma homocysteine (μ mol/L)	
	Mean $\pm$ SD	P-value	Mean $\pm$ SD	P-value
Controls	7.24 $\pm$ 1.05	(Ref)	15.31 $\pm$ 7.18	(Ref)
Luminal A	6.97 $\pm$ 0.96	0.046*	15.77 $\pm$ 8.56	0.62
Luminal B	6.89 $\pm$ 0.90	0.035*	18.29 $\pm$ 6.72	0.009*
LUMINA-HH	7.10 $\pm$ 1.20	0.46	11.27 $\pm$ 6.20	0.002*
LUMINB-HH	6.63 $\pm$ 0.95	0.05*	14.65 $\pm$ 9.92	0.67
HER only +ve	6.64 $\pm$ 1.08	<0.005*	13.00 $\pm$ 7.61	0.07
Basal-like	6.62 $\pm$ 1.34	0.0003*	17.73 $\pm$ 5.42	0.025*

levels in Luminal B and basal-like subtypes as observed in our current study could be due to MTHFR C677T variant in luminal B; and low plasma folate and low frequency of cSHMT C1420T variant in basal-like subtype. This is consistent with the observation of Pirouzpanah et al. [29] indicating inverse association of ER- α promoter hypermethylation with plasma folate and positive association with plasma homocysteine.

The tumor-node-metastasis (TNM) stage, the principal prognostic indicator currently being used in clinical practice for the breast cancer [30,31], was found to be associated by aberrations in one-carbon metabolism. The inverse association of tubule formation with cSHMT C1420T and MTHFR C677T variant alleles and positive association with RFC1 G80A variant allele as observed in this study, was consistent with the observation of Xu et al.

[32], showing reduced mortality in breast cancer patients harboring cSHMT C1420T and MTHFR C677T polymorphisms and increased mortality (although not statistically significant) in patients with RFC1 G80A polymorphism. The inverse association between the tumor grade and cSHMT C1420T can be explained based on the increased plasma folate due to induction of futile folate cycle by irreversible conversion of 5,10-methylene tetrahydrofolate to 5-formyl tetrahydrofolate [33]. The inverse association between the tumor grade and MTHFR C677T can be substantiated based on the study by Stankova et al. [34], demonstrating the beneficial effect of antisense-mediated inhibition of MTHFR expression in inhibiting the growth of human colon, lung, breast, prostate, and neuroblastoma tumor cells in vitro and human colon and lung carcinoma xenografts in vivo. The

Table 8. Association of Demographic and Genetic Factors With Breast Cancer Grade (TNM Classification)

Variable	Spearman rank correlation coefficient (Rho)		
	Tubule formation	Nuclear pleomorphism	Mitotic index
Age	-0.026	0.20***	0.25***
BMI	-0.005	0.005	-0.07
Age at menarche	-0.32***	0.11	0.04
Parity	-0.02	-0.09	0.04
Menopause	-0.07	0.07	0.12*
F/H of breast cancer	0.14*	0.25***	0.42***
GCPII C1561T	-0.05	-0.16**	-0.09
RFC1 G80A	0.19**	-0.09	0.11
cSHMT C1420T	-0.26***	-0.11	0.08
TYMS 5'-UTR 2R3R	-0.16**	0.08	0.23***
TYMS 3'-UTR del6/ins6	0.07	0.07	0.10
MTHFR C677T	-0.15*	-0.08	-0.11
MTR A2756G	0.12*	-0.05	0.19**
MTRR A66G	-0.09	-0.07	-0.14*
Sensitivity	0.85 (0.69-0.94)	0.63 (0.52-0.70)	
Specificity	0.82 (0.72-0.87)	0.87 (0.73-0.95)	

GCPII, glutamate carboxypeptidase II; RFC1, reduced folate carrier 1; cSHMT, cytosolic serine hydroxymethyltransferase; TYMS, thymidylate synthase; MTHFR, methylene tetrahydrofolate reductase; MTR, methionine synthase; MTRR, methionine synthase reductase.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

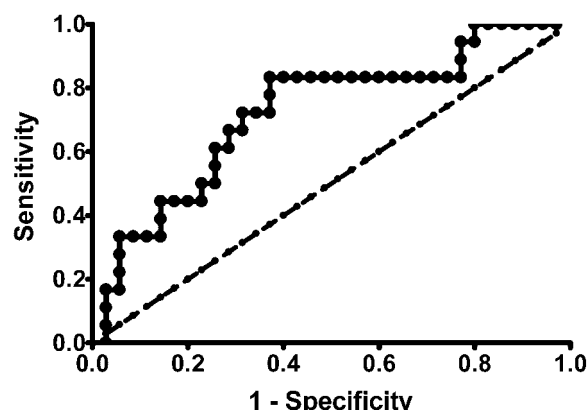


Figure 1. ROC curve for predicting breast cancer grade based on aberrations in one-carbon metabolism. Genetic polymorphisms i.e., GCPII C1561T, RFC1 G80A, cSHMT C1420T, TYMS 5'-UTR 28 bp tandem repeat, MTHFR C677T, MTR A2756G, and methionine synthase reductase (MTRR) A66G and physiological variables i.e., age, age at menarche, menopausal status, family history of breast cancer were used as predictor variables and tumor grade was used as outcome variable to obtain multiple linear regression based algorithm that can predict tumor grade. The sensitivity and specificity of prediction based on these factors was used for ROC curve plotting. Multiple linear regression models of aberrations in one-carbon metabolism were found to be moderate predictors of breast cancer grade (area under the ROC curve, $C = 0.72$, 95% CI: 0.58–0.87, $P = 0.008$). Dotted line indicates identity.

positive association between the tumor grade and RFC1 G80A variant allele can be explained based on the impaired transport of folate across the RBC membrane with the subsequent decrease in the red blood cell folate levels [35] and based on the observation that RFC1 is down-regulated under conditions of severe folate deprivation [36]. In vitro and in vivo studies showed that TYMS 5'-UTR 2R allele was associated low TYMS expression [37]. Nuclear pleomorphism showed positive association with age and familial breast cancer; and inverse association with GCPII C1561T polymorphism. Till date, no studies were conducted on this polymorphism directly with cancer progression. However, in vitro and preclinical targeted alpha therapy with Bi-213 labeled J59A antibody targeted against the prostate specific membrane antigen (an isoform of GCPII), was found to have in vitro cytotoxicity and in vivo anti-proliferative property [38]. Tubule formation, nuclear pleomorphism and mitotic index cumulatively can be moderately predicted by aberrations in one-carbon metabolism (area under the ROC curve = 0.72) as observed in the current study.

The strengths of the current study are novel findings of association between the aberrations in one-carbon metabolism and breast cancer subtypes that can help in understanding the mechanism of pathogenesis further. Since number of subjects in subgroups is relatively small, sample size calculations were done based on the estimated frequency of variant allele in two groups (phenotypic subgroup vs. controls) and all the associations had

power of $\geq 80\%$ and type I α -error of ≤ 0.05 . The possibility of some of the positive associations resulting from chance due to multiple testing although could not be ruled out completely, logistic regression based approach was used to control confounding effects of both demographic and genetic variables. Additional studies on large and ethnically diverse populations using micro-array based expression profiles are warranted to substantiate the role of aberrations in one-carbon metabolism as the prognostic factors for breast cancer. Future studies with additional biomarkers of cellular methylation such as S-adenosyl homocysteine and SAM might be beneficial in understanding the mechanistic aspects of these genotype–phenotype correlations.

To conclude, demographic characteristics, plasma folate, plasma homocysteine, and polymorphisms in GCPII, RFC1, cSHMT, TYMS, MTHFR, and MTR genes were associated with the clinicopathological characteristics of breast cancer. Differences in the distribution of these risk factors among subtypes of breast cancer might be useful in predicting the subtype of breast cancer as well as the disease progression. Multiple linear regression models of aberrations in one-carbon metabolism were found to be moderate predictors of breast cancer grade.

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